

Worked Answer Key

1. **P1-1:** 3:35; “thirty-five minutes past three.” **P1-2:** 8:45; “eight forty-five.” A quarter hour is 15 minutes, so “quarter to 9” means 15 minutes before 9.
2. **P2-1:** 12:30 a.m. is 30 minutes after midnight and before noon on the same day. **P2-2:** Add 12 to the hour for this p.m. time: $7+12=19$, so **19:45**.
3. **P3-1:** $2:35$ to $3:35 = 1$ h; to $4:10 = 35$ min; **1 h 35 min**. **P3-2:** $8:47$ to $9:00 = 13$ min; plus 32 = **45 min**.
4. **P4-1:** $11:40 + 20 =$ noon; $+35 =$ **12:35 p.m.** **P4-2:** $10:35$ to midnight = 1 h 25 min; $+1$ h 20 min = **2 h 45 min**.
5. **P5-1:** $3 \times 60 + 18 =$ **198 min**. **P5-2:** $250 = 4 \times 60 + 10$, so **4 h 10 min**.
6. **P6-1:** $9:55$ to $10:05 =$ **10 min**. **P6-2:** $9:10$ to $11:25 =$ **2 h 15 min**.
7. **P7-1:** 18 minutes before $7:10$ is $6:52$; **6:50 is earlier**. **P7-2:** **12:05 a.m., 11:50 a.m., 12:15 p.m., 11:50 p.m.**
8. **P8-1:** **\$7.06**, or **7.06 dollars**. **P8-2:** $75+20+20+7 =$ **122¢ = \$1.22**.
9. **P9-1:** $75¢+49¢=124¢$; regroup 100¢ as \$1, leaving 24¢. The dollars total $12+8+1=21$, so the sum is **\$21.24**. **P9-2:** $3.85+6.27 =$ **\$10.12**.
10. **P10-1:** $20.00-13.68 =$ **\$6.32**. **P10-2:** $35.50-18.75 =$ **\$16.75**.
11. **P11-1:** $10.00-7.46 =$ **\$2.54**. **P11-2:** $\$20-\$14.83 = \$5.17$; e.g., 5-dollar bill, dime, nickel, 2 pennies.
12. **P12-1:** $8.95+14.50+2.35 =$ **\$25.80**. **P12-2:** $12.49+9.75 = 22.24$; $30-22.24 =$ **\$7.76**; yes.
13. **P13-1:** $7.20 \div 4 =$ **\$1.80**. **P13-2:** $5.94 \div 6 =$ **\$0.99**.
14. **P14-1:** $5 \times 6.75 =$ **\$33.75**. **P14-2:** $3 \times 8.40 = 25.20$; $+2.25 =$ **\$27.45**.
15. **P15-1:** $27.60 \div 4 =$ **\$6.90**. **P15-2:** $45 \div 5 =$ **\$9.00**.
16. **P16-1:** $25-8.50-6.75 =$ **\$9.75**. **P16-2:** $8 \times 7.25 = 58$; $100-58 =$ **\$42**, enough.
17. **P17-1:** $40-8 =$ **\$32**. **P17-2:** tax = $12.50 \times 0.08 =$ **\$1.00**; final = **\$13.50**.
18. **P18-1:** $15 \times 6 =$ **\$90**. **P18-2:** $9:30 + 2$ h = $11:30$; $+45 =$ **12:15 p.m.**
19. **P19-1:** The correct change is $25-16.80 =$ **\$8.20**. The coins total $2 \times 25¢ + 3 \times 10¢ = 80¢$. The bill would need to be worth $\$8.20 - \$0.80 = \$7.40$. There is no U.S. bill worth \$7.40, so **the change is incorrect**. The exact amount given cannot be found without knowing the bill’s value. **P19-2:** $4:15 + 1$ h 35 min = **5:50 p.m.**
20. **P20-1:** $9:45$ to midnight = 2 h 15 min; $+20$ min = **2 h 35 min**. Cost $18.75+3.60 =$ **\$22.35**. **P20-2:** $2 \times 12.95 = 25.90$; $3 \times 4.25 = 12.75$; total 38.65; $50-38.65 =$ **\$11.35**.